

Jordan's Lemma

Standard complex-analysis result; treatment follows J. E. Santos' Cambridge IB lectures.

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Mathematical Methods

Statement. Let $f(z) = e^{iaz}g(z)$ with $a > 0$. Show that if $g(z) \rightarrow 0$ uniformly on the upper half-plane as $|z| \rightarrow \infty$, i.e.

$$\max_{\theta \in [0, \pi]} |g(Re^{i\theta})| \rightarrow 0 \quad \text{as } R \rightarrow \infty, \quad (1)$$

then the integral over the upper semicircular arc $\mathcal{C}_R = \{z = Re^{i\theta}, \theta \in [0, \pi]\}$ vanishes:

$$\lim_{R \rightarrow \infty} \int_{\mathcal{C}_R} f(z) dz = 0. \quad (2)$$

Find an upper bound for $\left| \int_{\mathcal{C}_R} f(z) dz \right|$ and give an example of a class of functions g satisfying (1).

Solution A typical example satisfying (1) is any rational function $g(z) = P(z)/Q(z)$ with $\deg Q \geq \deg P + 1$, for which $\max |g(Re^{i\theta})| = \mathcal{O}(R^{-1}) \rightarrow 0$.

Parametrise $z = Re^{i\theta}$, so $dz = iRe^{i\theta}d\theta$. The integral becomes

$$\int_{\mathcal{C}_R} f(z) dz = iR \int_0^\pi g(Re^{i\theta}) e^{iaRe^{i\theta}} e^{i\theta} d\theta. \quad (3)$$

Taking the absolute value and using $|e^{i\theta}| = 1$ and $|e^{iaRe^{i\theta}}| = e^{-aR \sin \theta}$:

$$\left| \int_{\mathcal{C}_R} f(z) dz \right| \quad (4)$$

$$\begin{aligned} &\leq R \int_0^\pi |g(Re^{i\theta})| e^{-aR \sin \theta} d\theta \\ &\leq M(R) \cdot R \int_0^\pi e^{-aR \sin \theta} d\theta, \end{aligned} \quad (5)$$

where $M(R) \equiv \max_{\theta \in [0, \pi]} |g(Re^{i\theta})|$. By symmetry of $\sin \theta$ about $\theta = \pi/2$:

$$R \int_0^\pi e^{-aR \sin \theta} d\theta = 2R \int_0^{\pi/2} e^{-aR \sin \theta} d\theta. \quad (6)$$

Since $\sin \theta$ is concave on $[0, \pi/2]$, it lies above the chord connecting $(0, 0)$ and $(\pi/2, 1)$:

$$\sin \theta \geq \frac{2\theta}{\pi}, \quad \theta \in [0, \frac{\pi}{2}]. \quad (7)$$

Therefore:

$$2R \int_0^{\pi/2} e^{-aR \sin \theta} d\theta \leq 2R \int_0^{\pi/2} e^{-2aR\theta/\pi} d\theta = \frac{\pi}{a} (1 - e^{-aR}) \leq \frac{\pi}{a}. \quad (8)$$

Combining, we obtain the bound

$$\left| \int_{\mathcal{C}_R} f(z) dz \right| \leq \frac{\pi}{a} M(R). \quad (9)$$

Taking $R \rightarrow \infty$: by assumption (1), $M(R) \rightarrow 0$, so the right-hand side of (9) vanishes by the squeeze theorem, and hence

$$\lim_{R \rightarrow \infty} \int_{\mathcal{C}_R} f(z) dz = 0. \quad (10)$$